# KLUSTER

## Build Guide & Technical Blueprint

Squadco Hackathon 2026 — Challenge 02: The Intelligent Economy

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**Team Setup:** 3 members | 1 Backend (Python/FastAPI) + 2 Frontend (React/Next.js)
**Timeline:** 5 days
**Challenge:** Design an intelligent economic platform connecting informal traders, job seekers, and financial services in one ecosystem

| 25% | 20% | 20% | 20% | 15% |
|---|---|---|---|---|
| Squad API Weight | Tech Architecture | Problem Insight | Econ. Viability | Presentation |

*+ 10% bonus for Impact Potential. A solution with no meaningful Squad API integration will not be eligible for top placements.*

# 1. The Core Idea

## What Is Kluster?

Kluster is an intelligent economic platform that digitizes **clusters** of informal workers — market associations, trade cooperatives, ajo groups, artisan guilds — rather than targeting isolated individuals. It uses transaction data flowing through Squad APIs to generate economic intelligence: cluster health scores, portable economic identities, demand-based opportunity detection, and data-driven financial product eligibility.

## The Core Insight

> The informal economy isn't broken — it's **invisible**. Ajo groups move billions. Market associations enforce contracts. Trade guilds train apprentices. These systems work. Kluster doesn't replace them — it makes them **legible** to the formal financial system without asking anyone to change their behavior.

## Why Clusters, Not Individuals?

Every other team will build for individual traders and individual job seekers. Kluster builds for **groups**. One market woman with no bank history is 'uncreditworthy.' But an ajo group of 30 women who've contributed weekly for years with a 96% contribution rate? That group is incredibly creditworthy — no bank just has a way to see it. Kluster is that visibility layer.

This also solves user acquisition: you don't convince 500 individuals. You convince one cluster leader who brings 500 members.

# 2. How It Works — End-to-End Flow

## Step 1: Cluster Onboarding

A cluster leader (market chairman, cooperative head) registers their group on Kluster. Members are added by phone number — no smartphone required. Each member automatically receives a **Squad virtual account**. This is their entry point into the digital economy.

## Step 2: Transactions Flow Through Squad

Customers pay into members' virtual accounts via bank transfer or USSD. Every payment triggers a **Squad webhook** that Kluster captures and logs — sender, receiver, amount, timestamp. The platform now has real economic activity data without anyone changing how they do business.

## Step 3: Economic Intelligence Emerges

Kluster computes a **Cluster Health Score** from aggregate transaction data (volume trends, active member rate, customer diversity, consistency). Each member also gets an individual **Activity Score**. An LLM generates narrative **Economic Profiles** from the raw data — portable, human-readable summaries of a person's economic life.

## Step 4: Demand Detection Routes Labor

When a cluster's transaction data shows growth signals — rising volume, new customer inflow, increased supplier purchases — the AI infers **labor demand**. Job seekers registered on the platform with matching skills and proximity are automatically surfaced as recommendations. No job postings needed. The economic activity *is* the signal.

## Step 5: Financial Products Unlock

Members whose individual score AND cluster health score pass thresholds are **pre-qualified** for financial products — micro-loans disbursed via Squad Transfer API, automated savings via recurring debits, group insurance pools. Loan amounts scale with both scores.

## Step 6: The Loop Compounds

Every transaction enriches the data. Scores become more accurate over time. Profiles deepen. Demand detection gets sharper. Workers who join clusters start transacting through Squad and build their own economic identity. The system demonstrably improves with scale.

# 3. Meeting the 5 Challenge Requirements

**Requirement 1**

**Identifies and digitally onboards informal traders and job seekers**

Cluster registration is the onboarding mechanism. A leader registers the group and adds members by phone number. Each member gets a Squad virtual account — their digital economic entry point. No smartphone, bank account, or business registration required. Job seekers register individually with skills, location, and language. The cluster model also solves distribution: one leader brings hundreds of members.

**Requirement 2**

**Matches job seekers to opportunities using AI (skills, location, language, economic context)**

Kluster doesn't use job postings. The AI monitors cluster transaction data and detects demand signals — rising volume, new customer inflow, seasonal spikes. It then routes job seekers with matching skills and proximity toward those growing clusters. **'Economic context'** isn't an afterthought — it's the primary matching signal, derived directly from Squad transaction data. A cluster whose revenue jumped 60% this month probably needs more hands. That's intelligent matching, not a job board.

**Requirement 3**

**Connects to financial services using alternative data and behavioural signals**

Alternative data = transaction patterns flowing through Squad virtual accounts (volume, consistency, customer diversity, growth trends) PLUS cluster-level signals (group health, member's importance within the group, collective track record). Financial services offered: micro-credit (disbursed via Squad Transfer API, repaid from incoming payments), savings (automated via Squad recurring debits), payments (all transactions flow through Squad). No bank statements, no payslips, no traditional credit history required.

**Requirement 4**

**Learns and improves over time as more data flows through**

Cluster health scores become more accurate with more transaction history. Economic profiles deepen as data grows — a profile based on 500 transactions tells a richer story than one based on 10. Demand detection gets sharper as seasonal patterns emerge across multiple cycles. Financial product eligibility thresholds can be calibrated based on repayment outcomes. Every Squad transaction makes every other function smarter. The data compounds.

**Requirement 5**

**Integrates Squad API as a core transactional or data layer — not optional**

Squad is simultaneously the **transactional layer** (virtual accounts receive payments, Transfer API disburses loans, recurring debits automate savings, USSD enables non-smartphone access, VAS adds ecosystem utility) AND the **data layer** (every transaction generates a webhook event that feeds the AI — scoring, profiling, demand detection all depend on this data). Without Squad, Kluster has no transactions and no intelligence. Squad IS the nervous system.

# 4. Satisfying the Four Pillars

## AI Automation

Cluster health scoring automates what a loan officer would take weeks to assess. Economic profile generation automates what would otherwise require a financial analyst. Demand-based opportunity routing automates what currently happens through word-of-mouth. All three are AI-driven processes that eliminate manual, painful work.

## Use of Data

Real-time transaction data from Squad webhooks powers predictive signals: growth trends, seasonal patterns, risk indicators, demand surges. The data isn't displayed in a dashboard — it drives automated decisions (scoring, matching, eligibility).

## Squad APIs

Virtual accounts (onboarding + data generation), Transfer API (loan disbursement, worker payments), webhooks (real-time AI triggers), USSD (accessibility), recurring debits (savings/repayment), VAS (ecosystem stickiness). Squad touches every core function — six integration points, not one.

## Financial Innovation

The innovation: using group-level economic data and network position as credit signals. A trader who supplies 15 other traders in a healthy cluster is creditworthy — that's a signal traditional scoring completely misses. Cluster-backed micro-credit, revenue-triggered financing, and group insurance pools are financial products designed for how the informal economy actually works.

# 5. Judging Criteria Mapping

| Criterion | Weight | How Kluster Addresses It |
|---|---|---|
| **Squad API Integration** | 25% | Six deep integration points: Virtual Accounts, Transfer API, Webhooks, USSD, Recurring Debits, |
| **Technical Architecture** | 20% | FastAPI + PostgreSQL backend, Squad API integrations, LLM-powered AI (profile generation + |
| **Problem Understanding & Innovation** | 20% | Cluster-first approach shows deep understanding that the informal economy runs on groups, not |
| **Economic Viability & Scalability** | 20% | Revenue: transaction fees, lending margins, data licensing. Scaling: 50 clusters x 200 members |
| **Presentation & Demo** | 15% | Live demo: cluster dashboard with economic graph, health scores, AI-generated profiles, deman |

| Criterion | Weight | How Kluster Addresses It |
| --- | --- | --- |
| **Impact Potential (Bonus)** | 10% | Millions of Nigerians already organized in informal cooperatives. Platform digitizes existing beha |

# 6. Squad API Integration Map

Every core platform function touches Squad. This section maps each Squad API to its role in Kluster's architecture.

| Squad API | Platform Function | What It Does in Kluster |
|---|---|---|
| **Virtual Accounts** | Onboarding + Data Generation | Every cluster member gets a unique virtual account on registration. Incoming payments cr |
| **Webhooks** | Real-time Data Capture | Every transaction fires a webhook event. Kluster logs it and updates cluster health scores, |
| **Transfer API** | Loan Disbursement + Worker Payments | When a member is pre-qualified for a micro-loan, the amount is disbursed directly to their |
| **Recurring Debits** | Savings + Loan Repayment | Automated weekly/monthly savings contributions and loan repayment schedules. Member |
| **USSD Payments** | Accessibility | Enables payments for users without smartphones or data. A trader can receive payment a |
| **VAS (Airtime, Bills)** | Ecosystem Stickiness | Users can buy airtime, data, pay electricity/cable from the platform. Adds daily utility and e |
| **Payment Links** | Trader Accessibility | Traders can generate a shareable link to receive payments via WhatsApp or SMS. No app |

# 7. The Dual Scoring System

Kluster uses two separate scores that work together. A member needs **both** to qualify for financial products. Strong individual in a dying cluster? Limited access. Weak individual riding a strong cluster? No free pass.

## Cluster Health Score

Measures group-level economic health from aggregate transaction data:

| Metric | What It Measures | Source |
|---|---|---|
| **Active Member Rate** | % of members with at least 1 transaction this period | Squad webhooks |
| **Volume Trend** | Is total cluster revenue growing or shrinking? | Squad webhooks |
| **Transaction Consistency** | Average regularity of payments across members | Squad webhooks |
| **Customer Diversity** | Number of unique external payers into the cluster | Squad webhooks |
| **Member Retention** | Are members staying active or going dormant? | Platform data |

## Member Activity Score

Measures individual economic performance:

| Metric | What It Measures | Source |
| --- | --- | --- |
| **Transaction Volume** | Total inflow over the scoring period | Squad webhooks |
| **Consistency** | How regularly do payments arrive? | Squad webhooks |
| **Customer Count** | Number of unique payers (repeat customers) | Squad webhooks |
| **Growth Trend** | Is revenue increasing or declining? | Squad webhooks |

## Loan Eligibility Formula

```
max_loan = base_amount x (individual_score / 100) x (cluster_score / 100)
```

Both scores must independently pass minimum thresholds before any financial product is offered. Poor performers aren't rejected — they see exactly which metrics need improvement and receive actionable guidance from the AI.

# 8. Demand-Based Opportunity Matching

**Key Design Decision:** Job seekers aren't matched to job postings. They're matched to **economic demand signals** detected from cluster transaction data. No one posts a job. The platform infers where labor is needed from how money is moving.

## How It Works

The AI monitors cluster-level transaction data for growth signals: rising volume, increased customer inflow, higher supplier purchases, growing weekly revenue. When these signals cross thresholds, the system infers labor demand and recommends workers with relevant skills and proximity.

## Example Scenario

A phone repair cluster in Computer Village shows rising transactions, more customer inflow, and increasing supplier purchases. Kluster detects rapid growth and recommends: repair assistants, apprentices, dispatch riders, and sales assistants — not because someone posted a job, but because the economic behavior suggests labor demand.

## Seasonal Detection

A tailoring cluster experiences increased activity before festive periods. Kluster predicts a temporary labor demand increase and recommends nearby tailors, apprentices, and delivery workers. The AI learns these seasonal patterns over time and pre-empts them in future cycles.

## The Worker Lifecycle

Once matched, workers start transacting through Squad and build their own economic identity. They gain repeat work, receive payments, and build an Activity Score. Eventually, they qualify for financial products themselves — or even form their own cluster. The job seeker becomes a full economic participant in the same system. **One unified loop, not a bolted-on job board.**

# 9. Technical Architecture (Backend Focus)

## Stack

| Layer | Technology | Why |
|---|---|---|
| **Backend** | Python / FastAPI | Fast to build, async support, great for API-first design |
| **Database** | PostgreSQL | Relational, supports aggregation queries needed for scoring |

| Layer | Technology | Why |
| --- | --- | --- |
| **AI / LLM** | Claude API (or similar) | Profile generation, demand matching explanations |
| **Payments** | Squad API (sandbox) | Virtual accounts, transfers, webhooks, USSD, VAS |
| **Frontend** | React / Next.js | Polished UI, data visualization capabilities |

## Core Database Tables

**clusters** — id, name, type (market_association / cooperative / ajo / guild), leader_id, location, description, created_at

**members** — id, cluster_id, name, phone, skills[], language, role, squad_virtual_account_id, is_job_seeker, created_at

**transactions** — id, member_id, cluster_id, amount, sender_ref, type (inflow/outflow), squad_transaction_ref, timestamp

**scores** — id, entity_type (cluster/member), entity_id, score, breakdown_json, calculated_at

**financial_products** — id, member_id, type (loan/savings), amount, status, squad_transfer_ref, created_at

**demand_signals** — id, cluster_id, signal_type, strength, recommended_skills[], detected_at

## Key API Endpoints (FastAPI)

```
POST /clusters — Create a new cluster
POST /clusters/{id}/members — Add members to a cluster
POST /webhooks/squad — Receive Squad webhook events (transaction logging)
GET /clusters/{id}/health — Get cluster health score + breakdown
GET /members/{id}/score — Get member activity score
GET /members/{id}/profile — Generate AI economic profile
GET /clusters/{id}/demand — Get detected demand signals
GET /matching/opportunities — Get demand-matched recommendations for a job seeker
POST /financial/prequalify/{member_id} — Check loan eligibility
POST /financial/disburse/{member_id} — Trigger Squad Transfer for loan
```

# 10. Five-Day Sprint Plan

| Day | Backend (You) | Frontend (2 Devs) |
|---|---|---|
| Day 1 | DB schema + migrations<br>FastAPI project scaffolding<br>Cluster & member CRUD<br>Squad sandbox account setup | Project scaffolding (Next.js)<br>Auth flow (if needed)<br>Cluster creation UI<br>Member management UI |
| Day 2 | Squad virtual account integration<br>Webhook handler endpoint<br>Transaction logging pipeline<br>Basic scoring queries | Cluster dashboard layout<br>Transaction list / feed<br>Member profile cards<br>Responsive layout |
| Day 3 | Cluster health scoring function<br>Member activity scoring<br>LLM integration (profile gen)<br>Demand signal detection logic | Economic activity visualization<br>(transaction flow diagram)<br>Health score display + breakdown<br>Demand signals UI |
| Day 4 | Opportunity matching endpoint<br>Loan eligibility logic<br>Squad Transfer API (disbursement)<br>Seed realistic demo data | Matching / recommendations UI<br>Loan eligibility display<br>Job seeker profile + onboarding<br>Polish all views |
| Day 5 | Bug fixes<br>End-to-end testing<br>Demo flow rehearsal<br>One-pager PDF | Final visual polish<br>Demo walkthrough testing<br>Pitch deck slides (10 max)<br>Rehearse presentation |

# 11. Demo Script (5 Minutes)

The demo tells one complete story through the platform. Every screen shown is functional, not a mockup. Pre-seed the database with realistic data for 2-3 clusters.

### 0:00 — 0:30 | The Problem

"47 million Nigerians work in the informal economy. They run real businesses, move real money, and employ real people. But to every bank and every institution, they're invisible. Not because they aren't productive — but because no system captures what they do."

### 0:30 — 1:30 | Meet the Cluster

Show the Oba Akran Phone Repair Association — a cluster of members collectively processing millions monthly. Show the cluster dashboard: transaction flow visualization, member cards, the health score at 82/100 with its breakdown.

### 1:30 — 2:30 | Economic Intelligence

Drill into a member's AI-generated economic profile — the narrative that turns raw transactions into a readable story. Show the demand signal: "This cluster's volume is up 60% — labor demand detected." Show recommended workers with skills and proximity matches.

### 2:30 — 3:30 | Financial Products

Show a pre-qualified member. Walk through the eligibility logic: individual score 88, cluster score 82, pre-qualified for a specific amount. Trigger the loan disbursement via Squad Transfer API — show the API response on screen.

### 3:30 — 4:30 | The Loop

"Every transaction makes the system smarter. Workers who join clusters build their own economic identity. Scores get more accurate. Demand detection gets sharper. The data compounds." Show a second, weaker cluster for contrast — lower scores, different recommendations.

### 4:30 — 5:00 | Scale

"50 clusters of 200 members = 10,000 users from 50 conversations. We don't acquire individuals — we onboard communities. Going national means partnering with market associations across Nigeria."

## 12. Scope Boundaries

**The demo is a vertical slice:** 2-3 pre-seeded clusters, a complete data flow from Squad transaction to AI insight to financial product. Small scope, complete story, every piece technically functional.

### We ARE building:

â■¢ Cluster and member management (CRUD + Squad virtual account assignment)

â■¢ Squad webhook ingestion and transaction logging

â■¢ Formula-based cluster health + member activity scoring

â■¢ LLM-powered economic profile generation

â■¢ Demand signal detection from transaction trends

â■¢ Opportunity matching (demand signals + job seeker profiles via LLM)

â■¢ Loan eligibility check + Squad Transfer API disbursement

â■¢ Polished React dashboard with data visualizations

â■¢ Seeded demo data that tells a compelling narrative

### We are NOT building:

â■¢ A graph database or complex network analysis algorithms

â■¢ ML models trained from scratch

â■¢ A mobile app

â■¢ Automated lending infrastructure or real financial products

â■¢ A real USSD interface (we describe the capability, demo shows web)

â■¢ Real user data — we use realistic synthetic data seeded for the demo

## 13. Submission Checklist

| Deliverable | Owner | Notes |
|---|---|---|
| **5-min product demo (live preferred)** | All | Pre-seed data, rehearse flow, have backup recording |
| **Pitch deck (max 10 slides)** | Frontend lead | Problem > User > Solution > Squad > AI > Flow > Impact > Scale > Validation |
| **GitHub repo (documented)** | Backend (you) | README with setup instructions, architecture notes, documented code |

| Deliverable | Owner | Notes |
|---|---|---|
| **One-pager PDF** | Backend (you) | Solution summary — write on Day 5 |
| **5-min Q&A** | All | Every team member must be able to speak. Prepare for scoring and Squad q |